

Fall 2025, EE 582: Assignment 1:
Introduction to Simulink and Modeling of Basic *RLC* Circuits
Due on Sunday, September 21 (submission in Canvas)

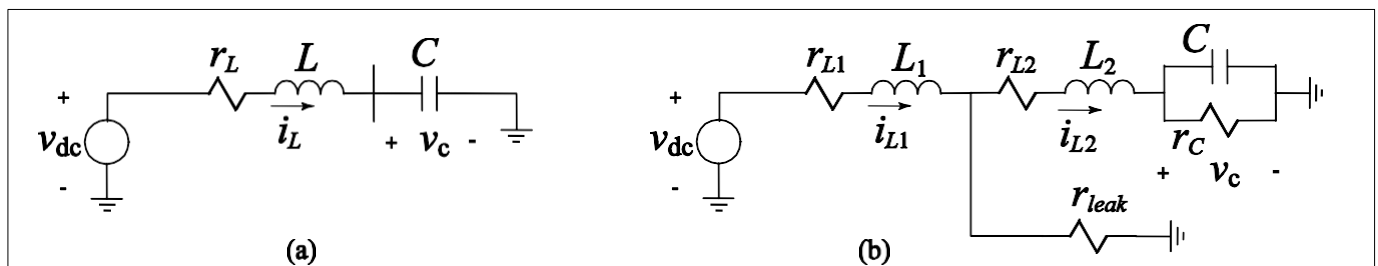


Fig. 1 A harmonic filter: (a) Simple model; (b) Model with more details.

SECTION A (Based on Fig. 1(a))

- 1) A shunt harmonic filter (series *RLC*) is shown in Fig. 1(a) above. It is designed for a three-phase 13.8kV 60Hz system to shunt the 7th harmonic current. Find the reactive power, damped and undamped resonance frequency, and quality factor of the filter. The circuit parameters are: $r_L = 1 \, \Omega$, $L = 19 \, \text{mH}$, and $C = 8.2 \, \mu\text{F}$.
- 2) Calculate the transient response of the circuit shown in Fig. 1(a) to a step voltage change if $v_{dc}(t) = s(t) = 11.3 \, \text{kV}$ for $t \geq 0$. Comment on the eigenvalues and response type.
- 3) Develop a Simulink model of the circuit shown in Fig. 1(a) using conventional Simulink blocks (i.e., summers, gains, integrals, etc.). Compute the transient response (solutions) on time interval $[0, 0.1] \, \text{s}$ using the ode3 solver with 0.1 ms time-step size.
- 4) Find the system eigenvalues using the MATLAB functions `linmod` and `eig` on the command line. Compare the results with part 2.
- 5) Considering the system eigenvalues, estimate the fixed-time step $h_{\max}$ that would be required by:
 - a) explicit Euler method (solver ode1); and b) explicit RK-5 method (solver ode5)?
- 6) Use any one of the fixed-time-step solvers of your own choice. Plot the solution for i_L and v_C . Experimentally establish the largest time step $h_{\max}$ that produces a stable and convergent solution that you can trust. Compare your result with the result of using the ode5 at $h = 0.5 \times 10^{-3} \, \text{s}$ and $h = 1.0 \times 10^{-3} \, \text{s}$.
- 7) Using a variable-time-step solver of your choice, establish a convergent solution that you can trust. Plot the solution i_L and v_C , and the time step h as functions of time. Your results should be accurate enough to be assumed as reference solution.
- 8) Using variable-step solvers, set $h_{\max} = 1 \times 10^{-2}$ and both error tolerances $\mathcal{E}_{abs} = \mathcal{E}_{rel} = 10^{-3}$. Experiment with all the available variable-step solvers. In a table, print the maximum time-step size $h_{\max}$ chosen by each solver and the total number of steps taken. Comment on how these solutions compare with the reference solution found in part 7. Which solver in your opinion is more efficient and does the best job for this system? Compare the step-size and the overall number of steps observed in parts 6, 7, and this part.

SECTION B (Based on Fig. 1(b))

- 9) In Fig. 1(a) the equivalent series resistance and inductance of the capacitor (r_{L2} and L_2) were added to the inductance parameters r_{L1} and L_1 (i.e. $r_L = r_{L1} + r_{L2}$ and $L = L_1 + L_2$). In Fig. 1(b), they are separated and also the capacitor leakage resistance (r_C) is considered. In addition, the discharge resistance (r_{leak}) is shown in Fig. 1(b). This resistance is used to discharge the filter capacitor when it is off-line for safety purposes which was ignored in the simple model shown in Fig. 1(a). Implement the model of Fig. 1(b) in Simulink using basic blocks considering $r_{L2} = 323 \text{ m}\Omega$, $L_2 = 1 \text{ mH}$, $r_C = 1 \text{ M}\Omega$, and $r_{leak} = 30 \text{ k}\Omega$. If the filter is disconnected, how long does it take for the capacitor to fully discharge?
- 10) Calculate the eigenvalues of the new system by hand and using MATLAB functions and repeat part 5 for the new model of Fig. 1(b). Is this system numerically stiff? Using any one of the fixed-time-step solvers of your choice simulate the model and plot the solution for i_L and v_C . Experimentally establish the largest time step $h_{\max}$ that produces a stable and convergent solution that you can trust.
- 11) Experiment with all the variable-step solvers using the setting of part 8. Comments on the results of using non-stiff and stiff solvers. Which solver do you recommend using? What setting gives you correct results? Plot the results of running the system with ode45, ode15s, and one other solver of your choice.
- 12) Discuss the effect of adding r_C and r_{leak} in the modeling accuracy and simulation numerical efficiency. Can we ignore any element with negligible errors to get a numerically efficient model which is easy to solve?

REPORTING

All reports should be typed. The assignment report should demonstrate the individual work and include printouts of the model windows, simulation results, as well as answers, discussions, and conclusions wherever appropriate. Use graphical commands such as **hold** and **subplot** for compact and clear presentation of the figures in your report. The discussions and comments should not be long (**avoid writing unnecessarily long reports!**), but they should demonstrate your understanding of the problem.

Best of luck!